Lifting Plan - Kobelco CKE90G-2

1. PLAN DETAILS

REFERENCE	ISSUED	PREPARED BY	CHECKED BY	APPROVED BY
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GROSS LOAD AT HOOK **2,700.00 kg**	CHART CAPACITY **16,000.00 kg** at this radius / boom angle	USAGE AT WORKING RADIUS **16.88%** **WITHIN LIMIT** of chart capacity
REQUIRED CAPACITY AT 75% **3,600.00 kg**	CRANE RATING **90,000.00 kg** checked against 75%	BOOM / RADIUS / ANGLE **40.00 m / 8.00 m / 48.00°**

Required capacity at 75%, W

$$W = \frac{\text{Gross load at hook}}{0.75} = \frac{2{,}700.00}{0.75} = 3{,}600.00 \text{ kg}$$

Usage, % of chart capacity

$$\text{Usage} = \frac{\text{Gross load at hook}}{\text{Chart capacity at radius}} \times 100\% = \frac{2{,}700.00}{16{,}000.00} \times 100\% = 16.88\%$$

2. LIFT INPUT PARAMETERS

SLING CONFIGURATION **3-Leg Sling in Each Leg of 2-Leg Sling Hitch (6 Effective Legs)**	SLING ASSEMBLY COUNT **2-sling bridle set**	SLING TOTAL LENGTH **7.0 m**
LOAD DIMENSIONS **Not required for the nested sling system**	DISTANCE BETWEEN PICK POINTS **5.4 m**	NO OF PILES **3**
SECOND SLING TOTAL LENGTH **0.600 m**	SECONDARY PICK SPACING **0.200 m**	SLING LEGS PER PILE **2**

3. GEOMETRIC & HEADROOM CALCULATIONS

Sling Contact Perimeter Wrap:

$$\text{Wrap Length} = 0.000 \text{ m}$$

Free Leg Length (First / Master Sling):

$$\text{Free Leg Length}_1 = 7.000 \text{ m}$$

Free Leg Length (Second / Secondary Sling):

$$\text{Free Leg Length}_2 = 0.600 \text{ m}$$

Transverse Sling Leg Angle from Vertical (Outer Secondary Legs 1 & 3):

$$\text{Angle}_{\text{Trans Outer}} = \sin^{-1}\left(\frac{\text{Offset}}{\text{Free Leg Length}_2}\right) = \sin^{-1}\left(\frac{0.200}{0.600}\right) = 19.47^\circ$$

Transverse Sling Leg Angle from Vertical (Middle Secondary Leg 2):

$$\text{Angle}_{\text{Trans Mid}} = \sin^{-1}\left(\frac{\text{Offset}}{\text{Free Leg Length}_2}\right) = \sin^{-1}\left(\frac{0.000}{0.600}\right) = 0.00^\circ$$

Rigging Arrangement & Pile Level Tolerance:
There are 3 piles in a row, so the master link hangs over the middle pile. The middle secondary leg is vertical and the outer pair slopes at a 0.2 m offset (Piles 1 & 3). All three legs are cut to the same free length and all three terminate on that one master link, so the pile tops cannot be level: the outer piles must stand higher than the middle pile by the amount derived below. The outer pair is the reference for the secondary height, because it carries the highest tension and gives the lowest angle factor.

Sling 2 Vertical Height (Top of Load to Master Link): (Governed by the outer sloped leg, Piles 1 & 3, the highest piles)

$$\text{Height}_{Sling2} = \text{Free Leg Length}_2 \times \cos(\text{Angle}_{Trans\ Outer}) = 0.600 \times 0.943 = 0.566 \text{ m}$$

Required Pile Level Difference (all legs cut to the same free length):

$$\text{Level Difference} = \text{Free Leg Length}_2 \times (\cos(\text{Angle}_{Trans\ Mid}) - \cos(\text{Angle}_{Trans\ Outer})) = 0.600 \times (1.000 - 0.943) = 0.034 \text{ m}$$

Total Combined Length for Longitudinal Calculation:

$$\text{Total Combined} = \text{Free Leg Length}_1 + \text{Height}_{Sling2} = 7.000 + 0.566 = 7.566 \text{ m}$$

Longitudinal Sling Leg Angle from Vertical:

$$\text{Angle}_{Long} = \sin^{-1}\left(\frac{\text{Master Spread}/2}{\text{Total Combined}}\right) = \sin^{-1}\left(\frac{2.700}{7.566}\right) = 20.91^\circ$$

Total Vertical Height (Hook to Top of RC Piles Headroom):

$$\text{Height}_{Hook-Load} = \text{Total Combined} \times \cos(20.91^\circ) = 7.566 \times 0.934 = 7.067 \text{ m}$$

4. RIGGING FACTORS & TENSION CALCULATIONS

Longitudinal Sling Angle Factor (AF Long):

$$\text{AF}_{Long} = \cos(20.91^\circ) = 0.934$$

Transverse Sling Angle Factor for Outer Legs (AF Trans Outer):

$$\text{AF}_{\text{Trans Outer}} = \cos(19.47^\circ) = 0.943$$

Combined Sling Angle Factor (AF) for Outer Secondary Legs:

$$\text{AF}_{\text{Combined Outer}} = \text{AF}_{Long} \times \text{AF}_{\text{Trans Outer}} = 0.934 \times 0.943 = 0.881$$

Transverse Sling Angle Factor for Mid Legs (AF Trans Mid):

$$\text{AF}_{\text{Trans Mid}} = \cos(0.00^\circ) = 1.000$$

Combined Sling Angle Factor (AF) for Middle Secondary Legs:

$$\text{AF}_{\text{Combined Mid}} = \text{AF}_{Long} \times \text{AF}_{\text{Trans Mid}} = 0.934 \times 1.000 = 0.934$$

Base Reeve Factor (RF):

$$\text{RF}_{Base} = 1.00$$

Tension in Each First Sling Leg (2 Master Legs Total):

$$\text{Tension}_1 = \frac{\text{Load Weight}}{\text{Sling Legs}_1 \times \text{AF}_{Long}} = \frac{2{,}700}{2 \times 0.934} = 1{,}445.4 \text{ kg}$$

Tension in the Sloped Outer Secondary Legs:

$$\text{Tension}_{\text{Secondary Outer}} = \frac{\text{Load Weight}}{\text{Sling Legs}_1 \times \text{Sling Legs}_2 \times \text{AF}_{\text{Combined Outer}}} = \frac{2{,}700}{2 \times 3 \times 0.881} = 510.8 \text{ kg}$$

Tension in the Straight Middle Secondary Legs:

$$\text{Tension}_{\text{Secondary Mid}} = \frac{\text{Load Weight}}{\text{Sling Legs}_1 \times \text{Sling Legs}_2 \times \text{AF}_{\text{Combined Mid}}} = \frac{2{,}700}{2 \times 3 \times 0.934} = 481.8 \text{ kg}$$

5. FINAL CAPACITY & EQUIPMENT SELECTION

Sling Leg Capacity Each for First Sling (Tag Rating):

$$\text{First Sling Leg Capacity Per} = \frac{\text{Installed Tag WLL}_1}{\text{Sling Legs}_1 \times \cos(\text{Tag WLL Angle}_1)} = \frac{5{,}000}{2 \times 0.707} = 3{,}536.1 \text{ kg}$$

Sling Leg Capacity Each for Second Sling (Tag Rating):

$$\text{Second Sling Leg Capacity Per} = \frac{\text{Installed Tag WLL}_2}{\text{Sling Legs}_2 \times \cos(\text{Tag WLL Angle}_2)} = \frac{3{,}000}{2 \times 0.707} = 2{,}121.6 \text{ kg}$$

Minimum Required Sling WLL for First Sling (Tag Rating):

$$\text{Minimum WLL}_1 = \frac{\text{Load Weight}}{\text{Sling Legs}_1 \times \text{RF} \times \text{AF}_{Long}} = \frac{2{,}700}{2 \times 1.00 \times 0.934} = 1{,}445.4 \text{ kg}$$

WLL Utilisation for First Sling (Tag Rating):

$$\text{WLL Utilisation}_1 = \frac{\text{Minimum WLL}_1}{\text{First Sling Leg Capacity Per}} \times 100\% = \frac{1{,}445.4}{3{,}536.1} \times 100\% = 40.88\%$$

Tag rating check (first sling): **PASS** — the sling leg is 40.88% utilised.

Minimum Required Sling WLL for Second Sling (Tag Rating):

$$\text{Minimum WLL}_2 = \frac{\text{Load Weight}}{\text{Sling Legs}_1 \times \text{Sling Legs}_2 \times \text{RF} \times \text{AF}_{Combined}} = \frac{2{,}700}{2 \times 3 \times 1.00 \times 0.881} = 510.8 \text{ kg}$$

WLL Utilisation for Second Sling (Tag Rating):

$$\text{WLL Utilisation}_2 = \frac{\text{Minimum WLL}_2}{\text{Second Sling Leg Capacity Per}} \times 100\% = \frac{510.8}{2{,}121.6} \times 100\% = 24.08\%$$

Tag rating check (second sling): **PASS** — the sling leg is 24.08% utilised.

Configured Setup SWL:

$$\text{Configured Setup SWL} = \text{Load Weight} = 2{,}700 \text{ kg}$$

6. NOTE

Choke Notes
No choke reduction applies to the selected arrangement.

To Calculate Sling Leg Capacity Each (Tag Rating):
If 4-Leg Sling, use Sling Legs = 3
If 3-Leg Sling, use Sling Legs = 2
If 2-Leg Sling, use Sling Legs = 2
Applied in this report: the first sling is a 2-leg sling, so the rule above gives Sling Legs = 2 for the first sling's leg capacity, and the second sling is a 3-leg sling, so the same rule gives Sling Legs = 2 for the second sling's leg capacity.

7. NONUNIFORM LOAD

Lift Input Parameters

Load weight	2,700.00 kg
Lift configuration	2-Leg Asymmetric (Shortening Grab Hook)
Standard sling length available (both legs)	7.000 m
Distance between pick points (total span)	5.000 m
COG horizontal position from pick point 1	3.500 m
Vertical height difference between pick points	0.500 m (pick point 2 higher)
Base reeve factor (RF)	1.00

8. GEOMETRIC & HEADROOM CALCULATIONS (SHORTENING METHOD WITH ELEVATION DELTA)

Operational rule: to make an asymmetrical lift without tilting the load, the crane hook must sit vertically above the centre of gravity. Because the COG is off-centre, one side of the horizontal span is shorter than the other. Both legs start at the same standard length, and the longer span keeps that full length and so fixes the hook's height. The pick points are not level: pick point 2 is 0.500 m higher than pick point 1. That step is not cosmetic. The long leg sets the hook's height above its own pick point, and the drop to the far pick point follows from the step, so raising it shortens that drop and lengthens the leg the operator has to shorten. The document states the rule as Y2 = Y1 - Height Difference; a positive entry means pick point 2 is higher and a negative one means it is lower.

Declared Horizontal Spans:
Left span (X1): 3.500 m

$$X_2 = \text{Total Span} - X_1 = 5.000 - 3.500 = 1.500 \text{ m}$$

Required Length for the Long Leg (Sling 1, Stays at Full Length):

$$\text{Length}_1 = \text{Standard Length} = 7.000 \text{ m}$$

Required Vertical Height from the Hook to Pick Point 1 Level (Y1):

$$Y_1 = \sqrt{\text{Length}^2 - X_1^2} = \sqrt{7.000^2 - 3.500^2} = 6.062 \text{ m}$$

Required Vertical Height from the Hook to Pick Point 2 Level (Y2):

$$Y_2 = Y_1 - \text{Height Difference} = 6.062 - 0.500 = 5.562 \text{ m}$$

Required Length for the Short Leg (Sling 2, Adjusted Leg):

$$\text{Length}_2 = \sqrt{Y_2^2 + X_2^2} = \sqrt{5.562^2 + 1.500^2} = 5.761 \text{ m}$$

Sling Shortening Adjustment Needed:

$$\text{Shortening} = \text{Standard Length} - \text{Required Length} = 7.000 - 5.761 = 1.239 \text{ m}$$

Sling Leg Angle from Vertical:

$$\text{Angle}_1 = \arcsin\frac{X_1}{\text{Length}_1} = \arcsin\frac{3.500}{7.000} = 30.00^\circ$$

$$\text{Angle}_2 = \arcsin\frac{X_2}{\text{Length}_2} = \arcsin\frac{1.500}{5.761} = 15.09^\circ$$

Sling 1 angle from vertical (3.500 m span)	30.00 °
Sling 2 angle from vertical	15.09 °

9. RIGGING FACTORS & TENSION CALCULATIONS

The load share is a statics result rather than a leg count: each leg carries the share set by the moment about the opposite pick point, so the leg on the short side of the COG carries the larger share. The angle factor then divides that share, because a leg standing further from vertical carries more tension for the same share.

Sling 1 Angle Factor (AF1):

$$\mathrm{AF}_1 = \cos(30.00^\circ) = 0.866$$

Sling 2 Angle Factor (AF2):

$$\mathrm{AF}_2 = \cos(15.09^\circ) = 0.966$$

Base Reeve Factor (RF):

$$\mathrm{RF} = 1.00$$

Load Weight Distribution to Pick Point 1:

$$\mathrm{Weight}_1 = \mathrm{Load} \times \frac{\mathrm{X}_2}{\mathrm{Total\ Span}} = 2{,}700.0 \times \frac{1.500}{5.000} = 810.0\ \mathrm{kg}$$

Load Weight Distribution to Pick Point 2:

$$\mathrm{Weight}_2 = \mathrm{Load} \times \frac{\mathrm{X}_1}{\mathrm{Total\ Span}} = 2{,}700.0 \times \frac{3.500}{5.000} = 1{,}890.0\ \mathrm{kg}$$

Tension in Sling 1 (Longer Leg):

$$\mathrm{Tension}_1 = \frac{\mathrm{Weight}_1}{\mathrm{AF}_1} = \frac{810.0}{0.866} = 935.3\ \mathrm{kg}$$

Tension in Sling 2 (Shorter Leg):

$$\mathrm{Tension}_2 = \frac{\mathrm{Weight}_2}{\mathrm{AF}_2} = \frac{1{,}890.0}{0.966} = 1{,}956.5\ \mathrm{kg}$$

Weight share, sling 1	810.0 kg
Weight share, sling 2	1,890.0 kg
Sling line tension, sling 1	935.3 kg
Sling line tension, sling 2	1,956.5 kg

10. FINAL CAPACITY & EQUIPMENT SELECTION

Minimum Required Sling WLL for Sling 1 (Tag Rating):

$$\mathrm{Min\ WLL}_1 = \frac{\mathrm{Weight}_1}{\mathrm{RF} \times \mathrm{AF}_1} = \frac{810.0}{1.00 \times 0.866} = 935.3\ \mathrm{kg}$$

Minimum Required Sling WLL for Sling 2 (Tag Rating):

$$\mathrm{Min\ WLL}_2 = \frac{\mathrm{Weight}_2}{\mathrm{RF} \times \mathrm{AF}_2} = \frac{1{,}890.0}{1.00 \times 0.966} = 1{,}956.5\ \mathrm{kg}$$

Governing Leg:

The governing leg is the one demanding the higher rated sling. It is the leg on the short side of the COG, because it carries the larger share and stands closer to vertical.

Minimum required sling WLL, sling 1	935.3 kg
Minimum required sling WLL, sling 2	1,956.5 kg
Governing leg	Sling 2 — 1,956.5 kg

Installed Tag Rating Check:

$$\mathrm{Capacity}_1 = \frac{\mathrm{Tag\ WLL}_1}{\mathrm{Sling\ Legs}_1 \times \cos(\mathrm{Tag\ Angle}_1)} = \frac{5{,}000.0}{2 \times 0.707} = 3{,}536.1\ \mathrm{kg}$$

$$\mathrm{Utilisation}_1 = \frac{\mathrm{Min\ WLL}_1}{\mathrm{Sling\ Leg\ Capacity}_1} \times 100\% = \frac{935.3}{3{,}536.1} \times 100\% = 26.45$$

$$\mathrm{Capacity}_2 = \frac{\mathrm{Tag\ WLL}_2}{\mathrm{Sling\ Legs}_2 \times \cos(\mathrm{Tag\ Angle}_2)} = \frac{6{,}000.0}{3 \times 0.866} = 2{,}309.5\ \mathrm{kg}$$

$$\mathrm{Utilisation}_2 = \frac{\mathrm{Min\ WLL}_2}{\mathrm{Sling\ Leg\ Capacity}_2} \times 100\% = \frac{1{,}956.5}{2{,}309.5} \times 100\% = 84.72$$

To Calculate Sling Leg Capacity Each (Tag Rating):

If 4-Leg Sling, use Sling Legs = 3

If 3-Leg Sling, use Sling Legs = 2

If 2-Leg Sling, use Sling Legs = 2

Applied in this report: Sling 1's tag is a 2-leg sling, so the rule gives Sling Legs = 2 for its leg capacity. Sling 2's tag is a 4-leg sling, so the rule gives Sling Legs = 3 for its leg capacity.

Sling 1 leg capacity	3,536.1 kg
Sling 1 utilisation	26.45 %
Sling 2 leg capacity	2,309.5 kg
Sling 2 utilisation	84.72 %

Configured Setup SWL:

$$\text{Setup SWL} = \text{Load Weight} = 2{,}700 \text{ kg}$$

11. TANDEM LIFT

Lift Input Parameters

Total load weight	2,700.00 kg
Lift configuration	Tandem Lift (Aligned COG Pivot)
Lug 2 position	Welded on the side or inside of the frame, inline with the horizontal COG axis.
Crane 2 projected distance, solved as	ND2 = D2 x cos(theta)
Sling arrangement	Straight vertical direct hitch — both sling legs vertical, so the angle factor is 1.000
Initial horizontal distance, crane 1 hook to COG (D-1)	3.000 m
Initial horizontal distance, crane 2 hook to COG (D-2)	5.000 m
Total span between crane hooks (L)	8.000 m
COG vertical depth below the lifting point baseline	2.000 m
Cargo tilt angle from horizontal	35.00°
Base reeve factor (RF)	1.00

The total span between the hooks is the sum of the two COG distances, because the centre of gravity lies between the hooks on the level baseline. The COG depth is the vertical distance from that baseline down to the physical centre of gravity inside the cargo, and it is what lets the tilt shift the COG sideways instead of only rotating the load about it.

The configuration decides how crane 2's distance is resolved once the load leans: ND2 = D2 x cos(theta) Stays wide and stable; the hooks do not have to migrate inward as the load rotates. Maximum rotation: Full 90 degrees vertical rotation. Use for: Up-ending and load rotation manoeuvres: precast concrete columns, vessels, storage tanks, wind tower sections.

12. GEOMETRIC & HEADROOM CALCULATIONS (DYNAMIC TILT METHOD)

Operational rule: the geometry is solved in order. The baseline 0° level lift fixes where the load divides between the two hooks, and the tilt is then applied, which shifts that division because the centre of gravity sits below the baseline joining the lifting points rather than in it.

Declared Sling Leg Angles:

$$\text{Angle}_1 = \text{Angle}_2 = 0.00^\circ \text{ from vertical}$$

New Horizontal Distance from Crane 1 to COG at 35° Tilt:

$$\text{ND}_1 = \text{h} \times \sin\theta + \text{D}_1 \times \cos\theta = 2.000 \times \sin(35.00^\circ) + 3.000 \times \cos(35.00^\circ) = 3.605 \text{ m}$$

New Horizontal Distance from Crane 2 to COG at 35° Tilt:

Lug 2 is inline with the centre of gravity, so it does not travel inward as the load leans; only the projection shortens it.

$$\text{ND}_2 = \text{D}_2 \times \cos\theta = 5.000 \times \cos(35.00^\circ) = 4.096 \text{ m}$$

New Total Effective Span Between Hooks (L-new):

$$\text{L}_{new} = \text{ND}_1 + \text{ND}_2 = 3.605 + 4.096 = 7.701 \text{ m}$$

Required Hook Height Differential to Hold the Declared Tilt:

$$\text{Height Difference} = \text{Total Span} \times \sin\theta = 8.000 \times \sin(35.00^\circ) = 4.589 \text{ m}$$

13. RIGGING FACTORS & TENSION CALCULATIONS

Each hook takes the share set by the moment about the other hook, so the crane nearer the COG carries the larger share. The angle factor then divides that share to give the sling line tension.

Sling Angle Factor (AF-1 and AF-2):

$$\text{AF}_1 = \text{AF}_2 = \cos(0.00^\circ) = 1.000$$

Base Reeve Factor (RF):

$$\text{RF} = 1.00$$

Case A — Weight Distribution on Crane 1 (0° Level):

$$\text{Weight}_1 = \text{Load} \times \frac{\text{D}_2}{\text{L}} = 2{,}700.0 \times \frac{5.000}{8.000} = 1{,}687.5 \text{ kg}$$

Case A — Weight Distribution on Crane 2 (0° Level):

$$\text{Weight}_2 = \text{Load} \times \frac{\text{D}_1}{\text{L}} = 2{,}700.0 \times \frac{3.000}{8.000} = 1{,}012.5 \text{ kg}$$

Case A — Sling Line Tension at 0° Level Lift:

$$\text{Tension}_1 = \frac{\text{Weight}_1}{\text{AF}_1} = \frac{1{,}687.5}{1.000} = 1{,}687.5 \text{ kg}$$

$$\text{Tension}_2 = \frac{\text{Weight}_2}{\text{AF}_2} = \frac{1{,}012.5}{1.000} = 1{,}012.5 \text{ kg}$$

Case B — Dynamic Weight Share on Crane 1 (35° Tilt):

$$\text{Weight}_1 = \text{Load} \times \frac{\text{ND}_2}{\text{L}_{new}} = 2{,}700.0 \times \frac{4.096}{7.701} = 1{,}436.1 \text{ kg}$$

Case B — Dynamic Weight Share on Crane 2 (35° Tilt):

$$\text{Weight}_2 = \text{Load} \times \frac{\text{ND}_1}{\text{L}_{new}} = 2{,}700.0 \times \frac{3.605}{7.701} = 1{,}263.9 \text{ kg}$$

Case B — Sling Line Tension at 35° Dynamic Slant Lift:

$$\text{Tension}_1 = \frac{\text{Weight}_1}{\text{AF}_1} = \frac{1{,}436.1}{1.000} = 1{,}436.1 \text{ kg}$$

$$\text{Tension}_2 = \frac{\text{Weight}_2}{\text{AF}_2} = \frac{1{,}263.9}{1.000} = 1{,}263.9 \text{ kg}$$

Case C — Weight Distribution at 90° Vertical Rotation:
Lug 2 sits under the COG on the level baseline, so the load can be rotated all the way to vertical without either hook passing under the COG. At 90° the COG is above crane 2's hook and beside crane 1's, which puts the whole weight on crane 2 and none on crane 1.

$$\text{Weight}_1 = 0 \text{ kg}, \qquad \text{Weight}_2 = \text{Load} = 2{,}700.0 \text{ kg}$$

$$\text{Tension}_1 = 0 \text{ kg}, \qquad \text{Tension}_2 = \frac{\text{Weight}_2}{\text{AF}_2} = \frac{2{,}700.0}{1.000} = 2{,}700.0 \text{ kg}$$

Tension, crane 1 (level)	1,687.5 kg
Tension, crane 2 (level)	1,012.5 kg
Tension, crane 1 (tilted)	1,436.1 kg
Tension, crane 2 (tilted)	1,263.9 kg
Tension, crane 1 (90° vertical)	0.0 kg
Tension, crane 2 (90° vertical)	2,700.0 kg

14. FINAL CAPACITY & EQUIPMENT SELECTION

Each crane must carry the highest of its own case tensions. The governing case is not the same crane in either direction, so each crane is quoted against the case that sizes it.

Minimum Required Sling WLL for Crane 1 (Left Hook Path):

$$\text{Min WLL}_1 = \max\left(\frac{\text{Weight}_{1,\,0°}}{1.00 \times 1.000}, \frac{\text{Weight}_{1,\,35°}}{1.00 \times 1.000}, \frac{\text{Weight}_{1,\,90°}}{1.00 \times 1.000}\right) = \max\left(\frac{1{,}687.5}{1.00 \times 1.000}, \frac{1{,}436.1}{1.00 \times 1.000}, \frac{0.0}{1.00 \times 1.000}\right) = 1{,}687.5 \text{ kg}$$

Governed by the highest value: the **0° level baseline** case governs crane 1, requiring 1,687.5 kg.

Minimum Required Sling WLL for Crane 2 (Right Hook Path):

$$\text{Min WLL}_2 = \max\left(\frac{\text{Weight}_{2,\,0°}}{1.00 \times 1.000}, \frac{\text{Weight}_{2,\,35°}}{1.00 \times 1.000}, \frac{\text{Weight}_{2,\,90°}}{1.00 \times 1.000}\right) = \max\left(\frac{1{,}012.5}{1.00 \times 1.000}, \frac{1{,}263.9}{1.00 \times 1.000}, \frac{2{,}700.0}{1.00 \times 1.000}\right) = 2{,}700.0 \text{ kg}$$

Governed by the highest value: the **90° vertical rotation** case governs crane 2, requiring 2,700.0 kg.

Governing Case Summary:

Minimum required sling WLL, crane 1	1,687.5 kg — governed by the 0° level baseline case
Minimum required sling WLL, crane 2	2,700.0 kg — governed by the 90° vertical rotation case
Governing crane	Crane 2 — 2,700.0 kg

Installed Sling Tag Rating Check:

$$\text{Capacity}_1 = \frac{\text{Tag WLL}_1}{\text{Sling Legs}_1 \times \cos(\text{Tag Angle}_1)} = \frac{25{,}000.0}{2 \times 0.707} = 17{,}680.3 \text{ kg}$$

$$\text{Utilisation}_1 = \frac{\text{Min WLL}_1}{\text{Sling Leg Capacity}_1} \times 100\% = \frac{1{,}687.5}{17{,}680.3} \times 100\% = 9.54$$

$$\text{Capacity}_2 = \frac{\text{Tag WLL}_2}{\text{Sling Legs}_2 \times \cos(\text{Tag Angle}_2)} = \frac{30{,}000.0}{2 \times 0.707} = 21{,}216.4 \text{ kg}$$

$$\text{Utilisation}_2 = \frac{\text{Min WLL}_2}{\text{Sling Leg Capacity}_2} \times 100\% = \frac{2{,}700.0}{21{,}216.4} \times 100\% = 12.73$$

To Calculate Sling Leg Capacity Each (Tag Rating):

If 4-Leg Sling, use Sling Legs = 3

If 3-Leg Sling, use Sling Legs = 2

If 2-Leg Sling, use Sling Legs = 2

Applied in this report: Crane 1's tag is a 2-leg sling, so the rule gives Sling Legs = 2 for its leg capacity. Crane 2's tag is a 3-leg sling, so the rule gives Sling Legs = 2 for its leg capacity.

Crane Rated Capacity Check (75% Allowable):

The rated capacity is the figure read from the crane's own load chart at the working radius, and it is the baseline this check is made against. Usage is the share of that charted capacity the hook path takes, so it is the demand over the charted figure and nothing else. At or below 75% the crane is within its allowable window; above the charted capacity it is over it.

$$\text{Usage}_1 = \frac{\text{Min WLL}_1}{\text{Rated Capacity}_1} \times 100\% = \frac{1{,}687.5}{50{,}000.0} \times 100\% = 3.38$$

Crane 1 is **WITHIN LIMIT** at 3.38 % of its charted capacity.

$$\text{Usage}_2 = \frac{\text{Min WLL}_2}{\text{Rated Capacity}_2} \times 100\% = \frac{2{,}700.0}{68{,}000.0} \times 100\% = 3.97$$

Crane 2 is **WITHIN LIMIT** at 3.97 % of its charted capacity.

Configured Setup SWL:

$$\text{Setup SWL} = \text{Total Load Weight} = 2{,}700 \text{ kg}$$

15. NOTE

Effect of the Declared Tilt on Each Hook

The same lift at the two declared load angles, for comparison. The level baseline is the more onerous case for crane 1 and the tilt is the more onerous case for crane 2; sizing either crane from the other case would understate it.

Load angle (tilt)	Crane 1 tension (left side)	Crane 2 tension (right side)	Hook height differential
0° (level lift)	1.69 t	1.01 t	0.000 m (level)
35° (slanted lift)	1.44 t	1.26 t	4.589 m (crane 1 higher)

Declared tilt direction: crane 1 lifts faster, so the cargo slants downwards towards crane 2. That transfers 251.4 kg onto crane 2 and releases the same weight from crane 1. The hooks must finish 4.589 m apart in height to hold the declared tilt. On this configuration the transfer continues if the rotation is carried past 35°: at 90° the whole 2,700 kg sits on crane 2.